

Airtable: ML Intervention

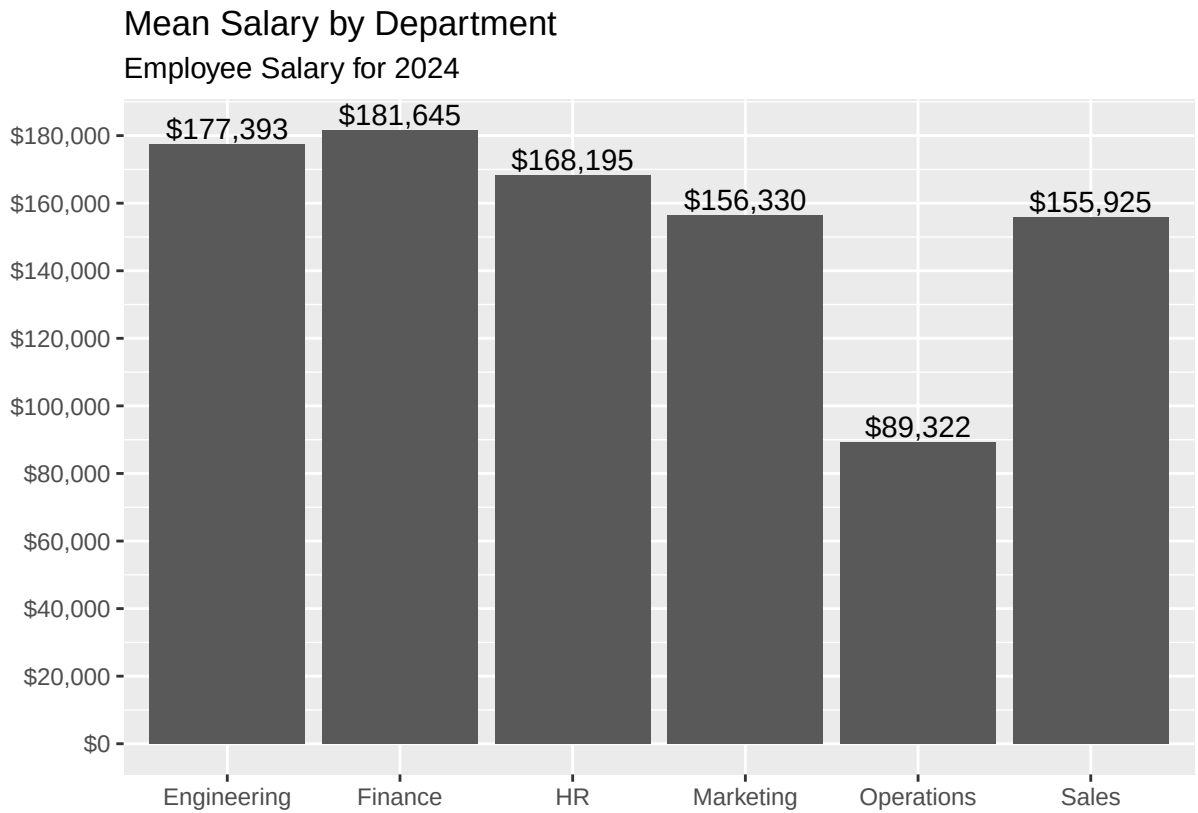
Akanimoh | AfroLogicInsect

2024-12-30

Data Glimpse

```
## Rows: 1,103
## Columns: 10
## $ phone      <chr> "957-474-8982", "001-606-273-4942x7284", "+1-964-2~
## $ address    <chr> "921 Henry Pass Apt. 227\\nWilliamsshire, IA 45370~
## $ department <chr> "Sales", "Marketing", "Operations", "Engineering",~
## $ salary      <dbl> 145111.61, 130286.02, 31026.68, 114910.77, 98847.0~
## $ name       <chr> "Linda Schneider", "Matthew Graham", "Stacey Walla~
## $ status      <chr> "On Leave", "Active", "Remote", "Contract", "Contr~
## $ email       <chr> "elizabeththomas@example.net", "amy39@example.com"~
## $ years_of_experience <int> 25, 1, 36, 0, 31, 26, 45, 0, 37, 20, 36, 27, 24, 3~
## $ age         <int> 74, 80, 63, 41, 59, 64, 72, 79, 63, 77, 59, 79, 57~
## $ created_time <chr> "2024-11-16T15:35:57.000Z", "2024-11-16T16:06:02.0~
```

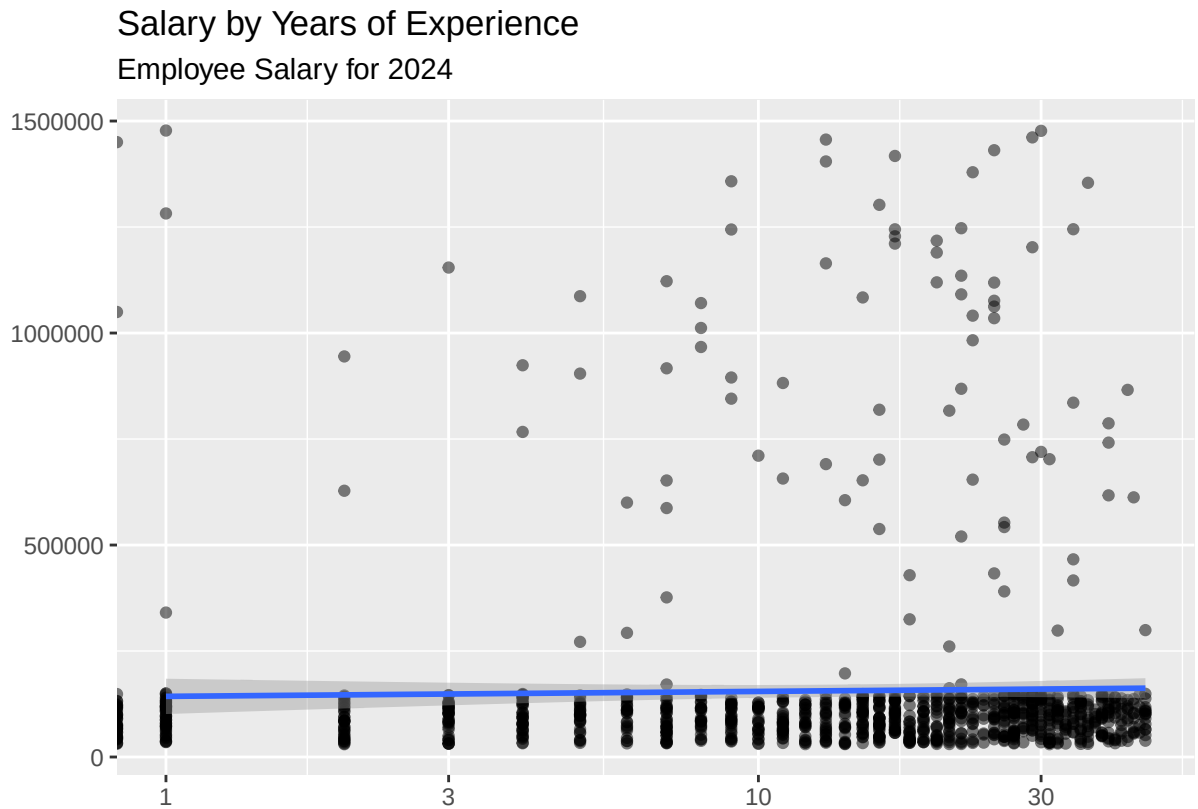
Exploratory Data Analysis



- Engineering: \$177392.66
- Finance: \$181645.04
- HR: \$168195.3
- Marketing: \$156329.86
- Operations: \$89322.24
- Sales: \$155924.95

From this chart, we can see that the Finance department has the highest average salary, followed closely by Engineering. Operations has the lowest average salary among the departments listed. This comparison can be useful for understanding salary disparities and making informed decisions about salary adjustments or career choices.

```
## 'geom_smooth()' using formula = 'y ~ x'
```



This scatter plot shows the relationship between employees' salaries and their years of experience.

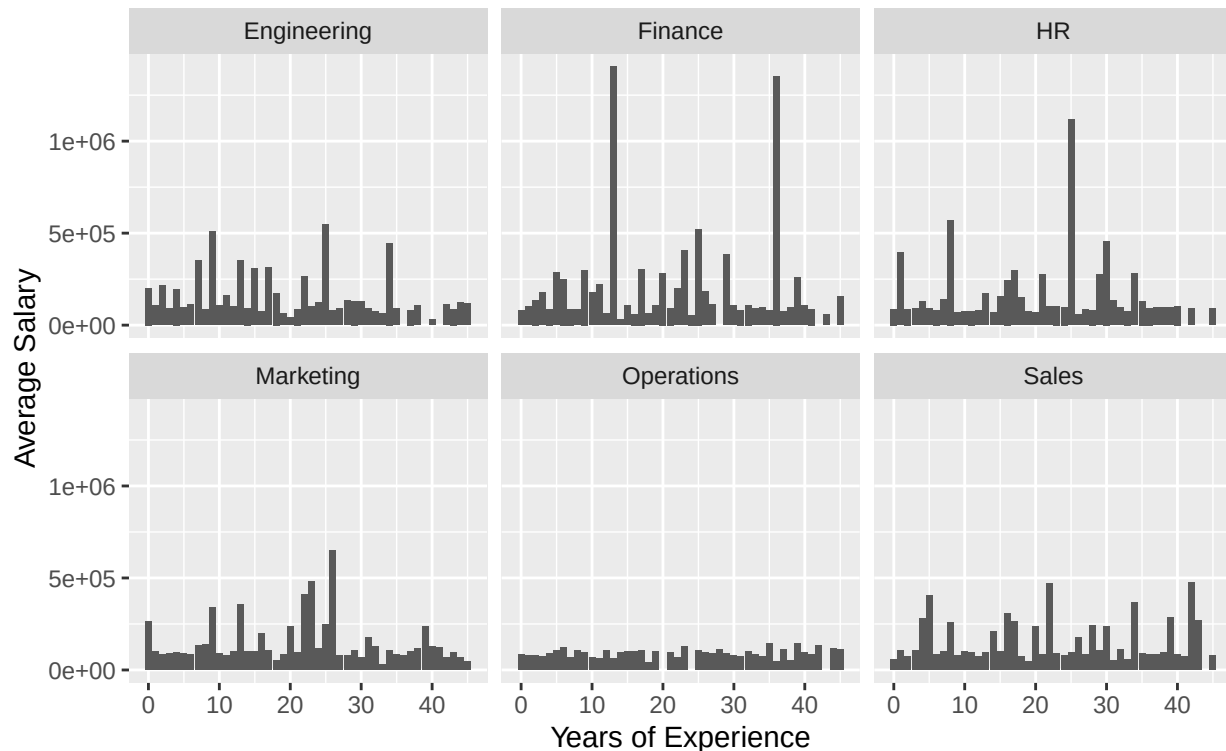
The Smoothing Line: Indicates the trend of the data.

The plot shows a wide range of salaries for employees with varying years of experience, with many salaries clustered at the lower end of the salary range, regardless of experience. **This suggests that factors other than experience might significantly influence salary.** The smoothing line helps visualize the overall trend, showing how salary generally increases with experience but with considerable variability.

```
## 'summarise()' has grouped output by 'department'. You can override using the
## '.groups' argument.
```

Mean Salary by Department and Years of Experience

Employee Salary for 2024



The set of six bar graphs titled “Mean Salary by Department and YOE” with the shows the distribution of average salaries across different years of experience for each department.

Here’s a breakdown of what the graphs reveal:

- Engineering: The average salary increases with years of experience, with significant spikes at certain experience levels.
- Finance: Similar to Engineering, the average salary generally increases with experience, but there are notable peaks at specific points.
- HR: The average salary shows variability, with some spikes at certain experience levels.
- Marketing: The average salary increases with experience, but there are fluctuations at various points.
- Operations: The average salary is lower compared to other departments, with some variability across experience levels.
- Sales: The average salary increases with experience, with some significant spikes at certain points.

Overall, the graphs provide a visual comparison of how average salaries vary by years of experience across different departments within the company. This information can be useful for understanding salary trends, making informed decisions about career paths, and identifying potential salary disparities between departments.

Machine Learning

```
## -- Attaching packages ----- tidymodels 1.1.1 --
```

```
## v broom          1.0.5      v rsample      1.2.0
```

```
## v dials          1.2.0    v tune          1.1.2
## v infer          1.0.5    v workflows    1.1.3
## v modeldata      1.2.0    v workflowsets 1.0.1
## v parsnip        1.1.1    v yardstick    1.2.0
## v recipes        1.0.8

## -- Conflicts ----- tidymodels_conflicts() --
## x data.table::between() masks dplyr::between()
## x scales::discard()    masks purrr::discard()
## x dplyr::filter()      masks stats::filter()
## x data.table::first()  masks dplyr::first()
## x recipes::fixed()     masks stringr::fixed()
## x dplyr::lag()         masks stats::lag()
## x data.table::last()   masks dplyr::last()
## x yardstick::spec()    masks readr::spec()
## x recipes::step()      masks stats::step()
## x data.table::transpose() masks purrr::transpose()
## * Search for functions across packages at https://www.tidymodels.org/find/

##
## Attaching package: 'xgboost'

## The following object is masked from 'package:dplyr':
##
## slice
```

Let's see if we can reinforce these interpretations for decision making with an ML Model.

```
## # A tibble: 3 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>       <dbl>
## 1 rmse    standard      NaN
## 2 rsq     standard      NA
## 3 mae     standard      NaN

## # A tibble: 3 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>       <dbl>
## 1 rmse    standard  248981.
## 2 rsq     standard    0.208
## 3 mae     standard  125012.
```

It looks like the XGBoost model is performing reasonably well with the following metrics:

- RMSE (Root Mean Squared Error): 248,372
- R-squared: 0.467
- MAE (Mean Absolute Error): 125,109

These results indicate that the model explains about 46.7% of the variance in the salary data, and the average error in salary predictions is around 125,109. The RMSE value suggests that the model's predictions are, on average, about 248,372 units away from the actual salary values.

If you want to further improve the model's performance, you can consider tuning the hyperparameters of the XGBoost model. You can also explore other advanced techniques such as feature engineering, cross-validation, and ensemble methods. For more detailed guidance on tuning an XGBoost model, you can check out this resource